INCH-POUND
MIL-S-13572C
AMENDMENT 1
25 April 1990

MILITARY SPECIFICATION

SPRINGS, HELICAL, COMPRESSION AND EXTENSION

This amendment forms a part of MIL-C-13572C, dated 9 March, 1988, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.1 Delete "MIL-L-3150 - Lubricating Oil, Preservative, Medium" and Add "MIL-C-16173 - Corrosion Preventive Compound, Solvent Cutback, Cold Application"

PAGE 3

2.1.1 Add "MIL-STD-1190 - Minimum Guidelines For Level C Preservation, Packing & Marking"

PAGE 4

3.1.1 Delete and substitute:

"3.1.1 Decarburization in Springs. Unless otherwise specified, prior to being wound material for Grade A spring shall be centerless ground to remove all surface decarburization and produce a true circular cross section. When magnified to 100 diameters there shall be no free ferrite visible. When specified (see 6.2), springs which are heat treated after coiling shall conform to the following depths of decarburization:

AMSC N/A FSC 5360

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TABLE 1-1. Maximum Decarburization in Grade A Springs [1]

	Hot Wound	Cold Wound
Complete[2]	0.000 in. (No complete Decarburization)	0.000 in. (No complete Decarburization)
Partial[3]	0.005 in.	1% Wire Diameter, not to exceed 0.0015 in.

[1] Measurements are made from the spring surface

[2] Complete Decarburization - loss of carbon content at the surface of a steel specimen to a level below the solubility limit of carbon in ferrite so that only ferrite is present.

[3] Partial Decarburization - loss of carbon content at the surface of a steel specimen to a level less than the bulk carbon content of the unaffected interior but greater than the room temperature solubility limit of carbon in ferrite."

PAGE 5

3.4.2 Delete and substitute:

"3.4.2 Grade A springs. Unless otherwise specified, steel springs shall be supplementary coated with a corrosion preventive compound conforming to MIL-C-16173 Grade 3 (P-3), or when specified on the drawing, phosphate coated in accordance with DOD-P-16232, Type M or Z, Class 1."

3.4.3 Delete and substitute:

"3.4.3 Grade B springs. Unless otherwise specified, carbon or alloy steel springs shall be cadmium plated in accordance with QQ-P-416, Type II, Class 2 or zinc coated in accordance with ASTM B633, Service Condition 3, Type II, except that the embrittlement relief requirements shall be in accordance with 3.4.4. When specified on the drawing, Type M or Z, class 1, phosphate coating per DOD-P-16232, and treated with corrosion preventive compound conforming to MIL-C-16173 Grade 3 (P-3), shall be used.

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PAGE 11

4.4.2 Delete and substitute:

"4.4.2 Decarburization. When centerless ground material grade A springs are required (see 3.1.1), one spring from each sample of springs, or two representative samples each 12 inches in length of the spring material that have been heat treated (samples of hot wound spring material have been heated prior to heat treatment) at the same time as the sample springs shall be examined for compliance with 3.1.1 by the following method: Samples of the material shall be cross sectioned, mounted in plastic, polished and etched with nitric acid and alcohol. In grade A springs, using a magnification of 100 diameters no free ferrite shall be visible. Failure to pass this test shall be cause for rejection of the lot represented by the sample."

PAGE 12

5.1.1, line 2: delete "drying procedure D-1 of".

5.1.2 Delete and substitute:

"5.1.2 Preservation. Application of a preservative is required only for alloy and carbon steel springs which do not have a permanent type coating such as paint, epoxy and platings, but including phosphate or black oxide. Unless otherwise specified, springs not exceeding three inches each in length or one pound each in weight shall receive an application of P-3 preservative of MIL-P-116. Unless otherwise specified, springs exceeding three inches each in length or one pound each in weight shall receive an application of P-1 or P-19 preservative specified in MIL-P-116."

PAGE 13

5.2.3 Delete and substitute:

"5.2.3 Level C. Cleaning, drying, preservation and unit packaging shall be in accordance with MIL-STD-1190."

PAGE 14

5.3.3 Delete and substitute:

"5.3.3 Level C. Packing shall be in accordance with MIL-STD-1190."

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NOTE: Marginal notations are not used in this amendment to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army-AR

Navy-OS

Air Force-82

Review activities:

Army-AT, AV, ME, MI

Navy-YD

DLA-IS

Preparing activity:

Army-AR

User activities:

Army-CR, ER, GL

Navy-AS, MC

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